Frictional Realities: Enabling Immersion in Mixed-Reality Performances

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ABSTRACT
This paper presents a case study of a Mixed-Reality Performance employing 360-degree video for a virtual reality experience. We repurpose the notions of friction to illustrate the different threads at which priming is enacted during the performance to create an immersive audience experience. We look at aspects of friction between the different layers of the Mixed-Reality Performance, namely: *temporal friction*, *friction between the physical and virtual presence* of the audience, and *friction between realities*. We argue that Mixed-Reality Performances that employ immersive technology, do not need to rely on its presumed immersive nature to make the performance an engaging or coherent experience. Immersion, in such performances, emerges from the audience' transition towards a more active role, and the creation of various fictional realities through frictions.

Author Keywords
Mixed reality performance; immersive video; immersion; priming; virtual reality; VR; interactive performance.

ACM Classification Keywords
• Human-centered computing~Mixed / augmented reality • Human-centered computing~Virtual reality • Human-centered computing~HCI theory, concepts and models • Human-centered computing~Collaborative interaction • Human-centered computing~Empirical studies in HCI • Applied computing~Performing arts • Applied computing~Media arts

INTRODUCTION
Over the past fifteen years, *Desert Rain* [30] has been the most emblematic example of a Mixed-Reality Performance (MRP) employing Virtual Reality (VR). As the technological barrier for using VR in performance art has drastically lowered, we are witnessing a renewed interest in the use of this technology. MRPs such as *Decompensation* [17], *Me and the Machine* [40], *Whist* [56], and trans-medial projects like *Orpheus* [52] provide examples of how artists adopt and appropriate 360-degree video.

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While VR and 360-videos are usually regarded as immersive technologies, in this paper, we argue that immersion is not an inherent quality of this medium that results from its technical properties. In so doing, we illustrate how artists enact performing strategies around the use of these media to create rich and meaningful audience experiences. We talk about these strategies as *frictions,* and we analyse how they are used within MRPs to create a dialogical relationship between the audience and performance, and create a meaningful, sensorial and embodied [38]. Within digital storytelling [2] frictions have been defined as inherent struggles in the plot that keep the audience engaged with the story. Here, we repurpose this notion to the domain of MRPs to discuss the audience participation to the narration of the performance, the fluctuation between the audience "real" (and physical) world and the "virtual" one of the 360-video, and between the physical and virtual presence of the audience in the performance. All these aspects of the audience experience –and not merely interacting with the technology- enable immersion with the 360-degree video and engagement with the performance as a whole. We regard the notion of friction as a "strong concept" [26] to talk about immersion and engagement within the application domain of Mixed-Reality Performances that use 360-degree video.

Of particular interest here are two related concepts of seamfulness and trajectories. Both concepts have broadly been explored and used in studies of mobile games and MRPs. Seamfulness explains how seams and gaps within the technology infrastructure can be used as features in the design of technology tools [5,11,15]. The concept of trajectories [8] describes instead the audience journeys through the performance, including narratives, spaces, times, roles and interfaces. Compared to seamful design that exposes glitches and transitions between one media to another, frictions are designed and enacted by the artists delicately to manipulate audiences' perception of time and space through the "live" experience. On the other hand, and compared to trajectories, friction shifts the focus from the audience journeys to blurring the boundaries between

audience and performers, and how this can result in a sensorial and embodied experience of the 360-degree video and the performance itself.

VR has been defined "as virtual space to be visited and navigated through" [41], and which users can interact with through various interaction modalities (i.e., lifting objects and experiencing their texture). Yet the common understanding of VR also includes more static videos (i.e., 360-degree videos) and using Head-Mounted Displays (HDMs) that block our view from the surrounding environment.

This paper introduces *Strange Days,* as an example of a Mixed-Reality Performance that uses 360-degree video and HMDs to virtually recreate the experience of past events, allowing audience members to visually experience them from the performers' point of view. *Strange Days* is the re-enactment of memories that the performers and the audience members share with each other during a live experience, designed and performed by Bombina Bombast [57]. In the tradition of TVX in investigating real-world experiences, studying *Strange Days* provided us with a valuable opportunity to explore the existing artistic practices [33] around VR use that contribute to an immersive and meaningful experience, and to illustrate the analytical and design issues such a focus raises.

We contribute by introducing frictions as a frame to understand immersion with 360- degree video in relation to the broader experience of participating in the performance. Immersion emerges from the audience's transition towards a more active role in the narration, emotional engagement with the story, and the tension between the sense of presence in the 360-video and in the physical space that includes other audience members and performers. The relevance of the contribution in this paper lays at the intersection of MRP and VR, and in the growing discourse on 360-degree storytelling [55]. We also contribute to the design of interactions with VR technologies that try to merge the virtual world with physical one (i.e., commercial systems such as Microsoft HoloLens and Facebook Social VR).

After reviewing related literature, we offer a detailed description of the performance we have studied, its setup, and the research method employed. We analyse the collected data applying the lens of frictions to highlight temporal friction, friction between the physical and virtual presence of the audience, and friction between realities. By reflecting on these findings, we discuss different methods by which immersion can be enabled.

RELATED WORK

Our case study is grounded in two main research areas: the HCI body of work on Mixed-Reality Performances, and research exploring the use of VR and immersive technologies in performing arts.

Virtual Reality in the Performing Arts

Over the past, few years there has been significant technological developments in virtual reality devices, in terms of their commercial potential, cost, and availability. Systems that used to have niche hardware requirement can now be supported with commercially available devices ranging from the HTC Vive VR headset to Google Cardboard that transforms smartphones into 'Mobile VR' headsets. The increasing number of projects creating content for different fields (for example games [45], education [22] and healthcare [44]), are evidence of growing interest in this field. HCI researchers have particularly focused on studies illustrating technical aspects of using virtual reality. These include challenges in augmented reality environments (such as depth perception [14]), tools (such as HMDs [1,16]) or novel techniques for optical configuration [25] in order to ease the visual discomfort [31], or exploring human behaviour addressing the impact of AR and VR experiences for chronic pain patients [24].

In *History of Virtual Reality Performance* [18], Dixon provided a comprehensive survey of virtual reality for performing arts up to 2006. Over a decade later, and with a considerable technological improvement of VR equipment (360-degree video recorders, haptic sensors, head-mounted display with high resolution and low motion blur, etc.), we have witnessed a new wave of curators, filmmakers, and artists using this technology to create artistic experiences. Yet, despite the popularity, there is little of academic work about VR in performances. Unlike the old VR performances that heavily relied on computer generated 3D environments [18], most recent productions use real world images and videos to create more photo-realistic [42] virtual reality environments. The key technology advancement that made this possible was the adoption of panoramic video cameras which allow for 360-degree video recordings. This comes at the cost of limiting the opportunities for interaction, as the videos are pre-recorded and offer little or no opportunities for the audience member to touch or manipulate content in the environment. Some performances rely on 360-degree video as the main component of the performance for the audience. A characteristic example is *Decompensation* [17], a Mixed-Reality Performance in which the artist recreates different psychological stages that a "refugee perhaps pass through when moving and settling into a new social or cultural context over the span of many years". Dehghani used 360-degree video to create an experience, in which participants can experience displacement by physically changing the location and (moving from one scene to another).

This renewed interest in VR for the performing arts calls for the development of research studies that help us understanding and analyse audience' engagement with VR.

Immersion

The term immersion is widely used in a number of fields such as theatre, art installation, performance art or VR

experiences. In theatre and performance art, immersion has been associated with audience participation and their intimate engagement with the piece or actors [35] as seen, for example, in promenade performances such as Sleep No More [58]. In these immersive performances audience becomes part of the performance and has the freedom (to some extent)[3] to interact with the performers, explore the space, and get engaged with performance activities [53]. In another reflection, several art installations (see for example *The Way Earthly Things Are Going* [49]) use the term to describe the sense of presence and how the art installation relies on one or multiple sensory inputs (i.e., hearing) to separate the spectator from "real world" and put her or him in an intimate moment in order to create an aesthetic experience. This exclusion from the "real world" (potentially) results in getting "immersed" in the artwork [35]. This definition, however, has been altered in those artistic practices that use digital technologies to immerse the audience [19]. In most of these digital immersive experiences, immersive technology (i.e., virtual reality) becomes the key feature to aid the immersion, often by recreating (or simulating) a lived experience for the audience. In this respect, Gordon Calleja [14] takes digital game studies as the main filed of exploration to highlight how the use of the term "immersion" varies from player's suspension of disbelief to player's perception of the designed world as well as players' engagement with the story, game world and characters. By this, he suggests the term immersion should be understood under the context it is used. In this regard and keeping digital games (that use interactive and 3-D designed environments) as the main context, Calleja looks at different level of player involvement with the game and how these involvements can enable an immersive game experience. While discussing these player involvements, he also looks at the immersion in three levels i) participation, ii) imagination, interpretation and interaction, and finally through player's iii) own narration of perceived experience.

Borrowing from Calleja's work on digital game studies, in this study we use the term immersion in the context of Mixed-Reality Performances as an interchange between the audience members' act of "absorbing" the art piece and artistic vision of MRP, and their transition towards a more active participation in the context of the performance [14].

Friction

Framing it within digital storytelling, Brayn Alexander defines friction as the inherent struggle in a plot in a story [2]. In this definition friction is not used with a negative connotation, rather it is the source of tension which grabs the attention of the audience, providing the energy that fuels their engagement beyond the curiosity sparked by interesting characters, style or world building. One example of this definition is different layers of mysteries and puzzles in a criminal story that the protagonist needs to uncover and solve in order to find the criminal (see for example, *The Killing* [59], a crime drama television series with twisted plots). While the complexity of the plot in this type of stories can keep the audience hooked to the stories, it should also have enough meaning for the audience to make sense and follow the story.

While Alexander sees frictions as a way to connect to the story, Stenros et al. [27] regard them as a source of disengagement. They associate the term with "issues" or "major cause of problems" in the narratives of a participatory drama. For Stenros et al. and their particular example of an alternate reality game, these problems can vary from character mismatches between online and live players, between fictional characters and production crew, or between aesthetics of online content and live events. They introduce these problems as frictions in narratives that disengages the player from the participatory drama and different beats of its story.

Having introduced these two different takes on the definition of friction, we adopt the term from Alexander's work to build upon the growing interest in VR and to discuss audience's engagement and immersion in the context of Mixed-Reality Performances that use 360-degree video.

Mixed-Reality Performance within HCI

Benford and Giannachi [7] introduced the term Mixed-Reality Performance to describe complex and hybrid artistic experiences which incorporate combinations of live and interactive performance where audience and performers interact with technology and digital media in real and virtual world. *Desert Rain* [6,30] is an emblematic Mixed-Reality Performance in which participants (players) experience virtual warfare in order to re-evaluate the boundaries between physical, fictional, real and virtual worlds. The "virtual world" in *Desert Rain*, is in fact, a combination of physical space and objects (designed for the purpose of the performance) and the projected video onto six rain curtains and screens. This setting was, however, different from those virtual environments that one can experience with recent VR technologies using HMDs. As a result of this performance and similar productions, the trajectories framework was developed [7,10]. In this analytical framework, performance experiences are viewed as journeys where the canonical narratives are intertwined with interpreted narratives, going through both spaces, times, roles and interfaces [8]. Several performance experiences have been designed using the trajectory framework, and researchers have illustrated how trajectories can be used in practice to analyse and control interactive performances and experiences [7]. This framework has been shown useful for both design and understanding of MRPs, such as enabling a dramaturgy for interactive experiences, identifying requirements for new technologies and for understanding better concepts within the overall performance experience [7,8,10].

Research on interactive performances have also investigated different modalities of audience participation such as audience's individual and social interaction with the artwork [16], audience's biodata as a source of input to interact with the performance [43] and audience's wittingness in relation

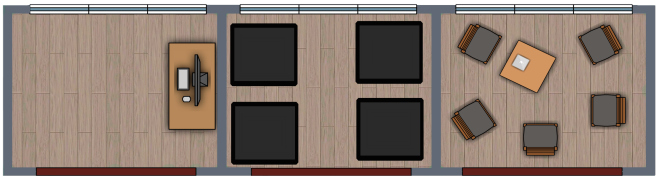


Figure 1. An illustration of the 3 rooms of the performance. Right: Room one, Middle: Room two, and Left: Room three.

to the performance [47]. The Humanacquarium [50,51], for instance, has focused on the mutual relationship between the performers and the audience and how participating can enable emotional encounters between different people and different roles.

In what follows, through analysis of a Mixed-Reality Performance we go beyond trajectories and seamful design to discuss how the notion of friction creates a dialogical relationship between the audience and performance in order to make the experience meaningful, sensorial and embodied. This relationship, we argue, enables immersion not only with the 360-degree video but the performance as a whole.

THE PERFORMANCE: STRANGE DAYS

Strange Days is a Mixed-Reality Performance that combines live performance with 360-degree video. It was designed and performed by Bombina Bombast, a leading performing arts company in Sweden that explores the use of innovative technologies (e.g., VR) in their work. Strange Days is inspired by a science fiction movie of the same name, directed by Kathryn Bigelow, in which the characters can re-experience the physical sensations of others. Bombina Bombast sought to realize this dystopian idea with currently available technology. The performance centres around the re-enactment of several memories and stories about important life events that the artists and the audience members share with each other. During the performance sharing is performed at three levels: through recounting the memories and the stories associated with them, through listening to songs associated with those memories, and through re-experiencing a memory using a 360-degree video.

Setup of "Strange Days"

This Mixed-Reality Performance is organized in three parts, each one taking place in a separate physical room (Figure 1). During the performance, audience members move through each room in sequence where they are met by a performer. When this study was carried out, audience members experienced the performance in groups of four; the rooms were fairly small (app. 10 m2) and located next to each other.

Room one

Strange Days starts with a facilitator leading participants into the first room (Figure 1, right). There, they meet the first performer, Steffan, who introduces himself and invites them to take a seat. On each chair there is a headphone set, all connected to a single iPad which is placed on the table and in front of the actor-performer (Figure 2 a). Stefan then sets out the narration by briefly introducing the performance and what they will experience in each room. Following this, he starts by sharing his own story and memory about his childhood, when he moved to Sweden from Serbia, during its civil war. He continues by describing how this ethnic childhood music suddenly became popular in the new country, and how the shared interest in this music genre helped him to connect to a new socio-cultural context. After this, he asks everyone to think about a song they value, then share the name of that song and the memories associated with it with the group. As audience members recall and retell their stories, he uses his tablet to find the song. Right after, he asks the audience to put the headphones on and listen to each song together, one by one. This part of the performance usually takes between 35-40 minutes, depending on the length of the songs and the conversation between the audience members and the performer.

Room two

In the second room (Figure 1, middle), audience members get invited by another performer, Emma, to sit or stand in the space that is covered by black rugs. These rugs are designed to resemble the floor in the 360-degree video that the audience members are going to explore. The performer then starts by sharing her story about a song that helped her face the grief and anguish of her parents' divorce. Then she describes an emotional and memorable day, when her partner surprised her by inviting a band to play that song just for her in their apartment. She gives a detailed account of that experience as she stood in the middle of the room, in the middle of the band which was playing for her, while her partner was holding her hand. Then she explicitly tells the audience that this is the memory that she wants to share with them.

After this line, she helps the audience put on the head-mounted displays and the headphones to watch and listen to the 360-degree video (Figure 2 b). In this video they experience the musical performance described from her point of view. A minute or two into the experience, the performer walks around the audience and holds each audience members' hand, one at a time, for 15-20 seconds. This is a way to share the special moment she experienced when her partner held her hand in the memory she recounts. The video is five minutes and 35 seconds long. The performance in the second room usually takes 10 up to13 minutes.

Room three

In the last room (Figure 1, left), Svante, the third performer, meets the audience and asks to choose who would receive a memory stick containing all the songs the group shared in the first room. After this, the performance ends and the audience members are led back to the waiting area.

RESEARCH METHODS

Over the last two years, *Strange Days* has been performed over 40 times in five different countries. However, this is, the first time that Bombina Bombast agreed to perform it for



Figure 2. (a) Performer (right) and audience members listening to each other's song (b) audience members viewing the musical performance in 360-degree video

research purposes. The artists' requirements were simply three small rooms located next to each other. The spatial setup was thus similar to the previous runs as the performances were designed and conceived to be portable. The layout of each room was organised to host four participants at a time, as well as one performer. Each group was scheduled to arrive at different times, thus minimizing the chance that they would meet and talk about the performance before experiencing it.

More than 300 people with different professional or amateur interest in performing arts, technology, design, or VR were invited, and heterogeneous group of 15 participants volunteered. They were 8 female and 7 male participants, ranging in age from 23 to 54 with a variety of professions including artists, researchers, students, developers, business directors and a retired teacher. We have given the participants pseudonyms to preserve their anonymity in relation to the quotes, however at the request of the artists we have used their real names throughout.

The purpose of our study was to explore the audience members' subjective experience of participating in the performance. This motivates our choice of data collection methods grounded in the social sciences (observations and interviews), drawing attention to the performance as an artefact (techniques and properties it embodies, see for instance [4]), or to the audience reception of the artistic vision.

Data collection

The empirical material was collected during four runs of the performance that took place on the same day. A total of four hours of observations was conducted while the participants experienced the performance. The observations focused on aspects of audience members' interactions with each other, and with the performing artists. We also looked at the participants' orientation towards the physical layout of the rooms in which the performance was experienced, and towards the other people present. For video recordings, one GoPro camera was placed in each room, which allowed us to capture an overview of the whole room, the people in it and the interactions between them. During the first run, one of the authors participated in the performance while observing the setting (participant observation). This was instrumental to gain a first-hand experience of the performance. To avoid being too intrusive, her interactions with the other audience members were limited to what was required by the artists.

When each performance was over, all the participants were interviewed individually through semi-structured interviews, in order to get immediate feedback from their experiences. Each interview lasted between 15-20 minutes and all recorded and transcribed. As the interviews were carried out immediately after the performance, they constituted the first occasion for the participants to discuss and share their experiences about the performance. We regard them as moments for the interviewees to utter their first impressions, and to start developing an opinion about the whole experience through talking to somebody else. We asked participants to describe their experience with the 360-degree video, and their experience in the different rooms, including their connection to the story or their contribution to the performance through sharing their personal memories and stories.

The authors of this paper were not involved in design of the performance nor its deployment. Our only contribution was to invite audience members and schedule their arrivals, and provide space and equipment for Bombina Bombast to reform and set the performance. In this respect, artists were not involved in designing the interview questions, collecting and analysing the data or suggesting anyone to participate in this study. While this collaboration provided us an opportunity to study existing MRP, for the artists it was a chance to advertise their work and performing in yet another venue. Artists did not receive any monetary compensation for this collaboration.

Data Analysis

The data were analysed recursively following principles of thematic analysis [13]. The authors started by collaboratively coding the transcribed interviews, focusing on participants' reflections on the experience of the 360-degree video as well as of their engagement with the performance as a whole. Being together vs. being alone in the experience, feeling uncomfortable, presence in physical and virtual rooms, re-experiencing and re-living, active vs. passive participation, physical awareness and encountering others and interacting with others (i.e., eye contact) were some of the initial themes identified during this first round of analysis.

During a second iteration, we turned to the notion of friction as a source of audience engagement [2] and we mapped it to the elements of the performance that we had identified as contributing (or not) to the i) audience involvement in the performance and ii) how they made sense of the 360-degree video in the context of the performance as a whole. This resulted in the three frictions that we present in this paper.

The video material was also used to analyse the moments of interactions that the participants had mentioned during the interview. This included, for instance, motivations for sharing a memory and how this was concretely instantiated in the narration, or verbal and non-verbal interactions between audience members. The video recordings offered an opportunity to examine how interactions between the artist-performers, the audience and the technology played out, and

what challenges and opportunities these interactions provided for the experience [25,34].

RESULTS

Repurposing the notion of friction to be applicable for MRPs is relevant as it provides a key terminology to talk about immersion as emerging from audience's engagement with the whole performance and not merely with the VR. Our results span three aspects of friction that enable audience engagement with the performance. These frictions are explained below with representative quotes from our participants, describing their experience of time and memories, their virtual and real presence, and the different realities they experienced. We need to indicate that the quotes presented below stand for the majority of participants who discussed the relevant topic except two quotes that illustrate different experiences by two participants.

Therefore, the three frictions introduced here, are representative and inclusive of the whole corpus of data. Differences can however be found in the audience experience of these frictions as we describe them in relation to each friction.

Temporal Friction

The first instance of friction we discuss is the temporal friction. This is an invitation to the audience members to move towards a more active role in the performance and to directly participate with their own stories. In the case of the temporal friction, this transition occurs alongside a temporal dimension, as the present narration in the performance intertwines with the audience members' past experiences.

Meeting one of the performers in the first room (Figure 2 a) marks the audience's initial experience of the performance. As everyone sits around the same table, the artist-performer begins the narration by sharing a story about his youth and the importance of music during this period of life. The quote below, extracted from the dialogues in the first room, provides a concrete example of how the artist-performer relates to music to unpack the emotional dimension of past events.

"...It was quite hard at that time to be Serbian in this country, and I listened a lot to Balkan music…it became central for my life, early teenage life, which was kind of the only thing that I had, which was mine, and I was so proud of this, even though it was so sad and so much war was going on…" (Steffan)



Figure 3. Inviting the audience members to take part and share their memories

With these words, the artist-performer ends the personal account of past experience. Right after, he addresses the audience members and explicitly asks them to take the same retrospective journey through recounting their own memories and associating a song to them. The following quote is extracted from the video recordings of the part of performance described here:

"…now I'd like you to think of one song that has mattered sometime in your life, you can start by thinking of a band, and then maybe a song and what [it] makes you remember of a specific moment when you listen to that song. Or something happened…". (Figure 3)

We regard this invitation to directly participate and contribute to the performance as a friction that makes the audience members move towards a more active role in the performance. The quote below is transcribed from one of the video recordings in the first room, and it provides an example of how an audience member shared her song with the other people present. This act places her story in tandem with the performer's story.

"…the reason I chose this song, because …still today when I hear it, it brings me back to those memories, kind of nostalgia. And it was the first time I actually had a song with someone, to say 'this is our song'." (Fera)

In this specific instance, the transition towards a more active participation modality occurs alongside a temporal dimension. Here the audience's present experience of the performance, their retrospection of meaningful past events, and the narrativization of those events (i.e., recounting them for other people) *come together* in the performance. In this process not only the performers', but also the audience members' pasts become central elements of the narratives and performance.

Different from temporal trajectories that looks into story lines designed by the artists and clock time in a real world [8], *temporal frictions blur the boundaries between audience and performers, thus allowing for a gradient of connection and engagement between the audience, the artist performer, and the performance as a whole.* The possibility to contribute with their own stories delicately set the audience members in the centre of the story, while turning the performance into a more intimate and *meaningful* experience.

The transition towards a more active participation was experienced as unexpected by all, but not necessarily uncomfortable [9]. It also contributed to a degree of self-consciousness in choosing and sharing certain stories with other people:

"It was unexpected, and you become a bit self-conscious when you need to choose something and share, and when you're not prepared". (Anna)

Resonating with what said by 12 participants, this quote also illustrates that the experience of a more active form of spectatorship intertwined with the audience members' awareness of their visibility in the performance.

"I was just thinking while the others were talking. Also trying to listen to them. It's also like you had to conduct how to tell the story. It was like the spotlight is on you. Go, tell your story" (Andrew).

Friction between Realties

The second type of friction we introduce is the one between the fictional realities emerging from the audience' experience of the performance and their experience of the "actual" world (i.e., the one they are located in). The adjective fictional simply indicates that it is part of the narrative [20]. This friction relates to the tension stemming from the audience projection into the fictional reality of the performance, both when memories and songs are shared, and when Emma's memory is replayed in the 360-degree video.

In the first room, collectively listening to the songs gave audience members the opportunity to empathize with the others, and to imagine them through the memories retold. As seen above, this was an unproblematic aspect of the audience engagement. As put by one of the interviewees, the memories retold were believable in the context of the performance, even if the events had not actually taken place. Listening together to the chosen songs was an opportunity to imagine the others, and see oneself in the situations recounted:

"When I listened to the songs I imagined them. Like, one guy was telling a story about when he was doing a road trip in America and I imagined him driving a car and I imagined another guy dancing. And I saw myself in my own songs." (Jason)

In the second room, however, the projection was not straightforward. Here, the 360-degree video provided a concrete visual layer to the performer's memory, as audience members could experience the physical layout of the room, and be surrounded by the band playing for Emma. By wearing the HMD, they could put themselves in Emma's original position in the room and (re-)experience the event from "her point of view":

"Now I would like to share this memory with you, and welcome you into it"

Narratology studies have extensively discussed the "suspension of disbelief" [20] as the agreement whereby readers (spectators in our case) pretend that the imaginary story being told has actually taken place and, thus, accept to believe in it. What is relevant for our analysis of a Mixed-Reality Performance is the role that digital media can play in suspending disbelief, or not.

Most of the interviewees (9 out 15) described their experience with the 360-degree video and with being present in someone's memory as strong and intimate. These people were also aware that their engagement with the story was not only connected to 360-degree video, but also to the story retold by Emma before they put on the HMD. The quote below represents this aspect of the audience' experience as they make sense of the relationships between Emma, the room and the song played, thus adding meaning to the content of the 360-degree video:

"It felt really private that you in a way could re-experience that memory. Strange to be present in someone else's memory because you can hear a memory in a story or someone telling you a memory, but this became another step. To be present, and get immersed in that because you have heard the story that she told me before. It became like I was there". (Tom)

Two of the interviewees reported a different experience with 360-degree video particularly regarding their projection with the fictional reality the medium was supposed to contribute to. One issue was the emotional disconnection between their personal past experience and the performer's (Emma) life.

"We were the [performer]character. In their past there was a song that would evoke an emotion of a life altering event, parents' divorce I think, without having any notion of that emotion there's a huge disconnect." (Verner)

Another disconnection stemmed from the technology, with respect to the technical properties and to people's concerns about what the technology could be used for (i.e., creating false memories). In the first quote below, Hanna explains that she did not experience any immersion in the 360-degree video because she is taller than the character in the video, that is the height of the camera when the video was recorded. This created a tension between the perception of her own body and presence in the actual world, and the first-person perspective mediated by the video in the virtual space. Hanna describes this experience as problematic from a technical perspective:

"I didn't like the immersion, I mean for the VR immersion is very important, I think immersion and the presence are not so good because I think the person is kind of smaller than the normal size, so the perspective is kind of weird."(Hanna)

The second quote below illustrates instead the disconnection between the audience' personal experience of the actual world and the fictional reality as determined by a meta-reflection on the role of technology:

"I felt like someone was trying to transplant memories into me …it would be really creepy and just super weird. Actually being able to relive somebody's memory,…that would be a really powerful and psychedelic experience. I can imagine that'd be very dangerous." (Verner)



Figure 4. (a, b) One of the audience members is trying different angles of viewing 360-degree video (c, d) Embodied interaction between the performer and audience

In the following section we discuss how this disconnection was lessened by bringing back the sense of presence, through performer's interaction with the audience.

Friction between the Physical and Virtual Presence

The third friction we introduce is the one between the audience's physical and virtual presence in the performance. Discussing the audience members' sensorial experiences during the performance, and the awareness of their bodies, underlies this friction.

The audience experience of the performance was strongly influenced by their sense of presence in the physical room and in the virtual space reproduced by the 360-degree video. In the first room, audience members interacted with each other by recounting personal stories and memories. This created an intimate, sometimes emotional or awkward situation among a group of people who did not necessarily know each other. The quote below represents this aspect experienced by all audience members:

"I think it was a bit awkward because it was so close and I didn't know people" (Mary).

The physical proximity of others amplified the audience members' awareness of their presence in the room, their body movements, and how they physically oriented themselves towards the other people's present. This physical awareness and presence were even more intense when everybody listened to the song that was being shared, as audience members became aware to be at the center of the other people's attention:

"…Just the fact that they were there and the fact that I shared my story with them, it changed it quite obviously and their presence made me feel self-conscious. I was aware of my body for example. I was aware of whether I'm tapping my feet or where I'm looking. Just because of the other people that were there. Totally different experience to sitting by yourself at home and listening to music" (Verner)

In the second room, however, this sense of self-awareness and presence weakened as audience members put on the HMDs to revisit Emma's memory. The 360-degree video enabled the audience to experience a virtual place (the one represented in the video) which they could inhabit individually. Wearing the HMDs and headphones allowed audience members to isolate themselves from the sight and sound of the surrounding physical environment, and thus to be less concerned about the presence of other people and how they experience or influence the same performance. 11 participants described this aspect of the experience during the interviews. During the performance we could observe, for instance, that members of the audience took small steps, moved around the room, bent to explore the boundaries of the 360-degree video, or simply sat on the floor to experience the video (Figure 4 a,b).

The moment in the performance when Emma holds the hand of each audience member connects the audience's virtual presence in the immersive video to the physical world they inhabit. This moment is emblematic of the friction between the physical and virtual presence of the audience and performer. Emma tried to replicate the scene and the sensorial experience of her partner holding her hand, by establishing direct physical contact with the audience. When Emma reminds the audience of their physical presence, by touching their bodies and holding their hands, she adds this sensorial perception to what they see in the 360-degree video.

"…when she held my hand, I felt like I don't want to leave her hand…It was a very good feeling. I felt that I'm more connected to this experience. Somehow, I felt it might be the thing she [Emma] experienced for real… [he] grabbed her hand and this is actually what she experienced." (Nina)

Emma's *"ghost hand",* as one of the participants put it, resulted in an intimate moment for the audience, while she was virtually present but physically invisible to her presents. This contributed to blurring the boundaries between the physical and the virtual space and presence. These particular moments can be seen in Figure 4 (c,d), when audience members react to performer holding their hands by physically turning their head to her side, and trying to find her in the 360-degree video.

DISCUSSION

Immersive videos and VR technology provide a range of creative opportunities for MRPs (e.g., enabling the audience to re-experience a memory from the performer's point of view). Throughout this paper, we have argued that the audience immersion with the 360-degree videos, and engagement with the performance as a whole is not a given property of the video as a medium. This is relevant for the TVX community, as a growing number of academic (i.e., [12,36]) and industry work (i.e., [60]) explores the role of VR for entrainment as well as artistic purposes.

In the analysis of Strange Days, we have illustrated how the audience immersion and engagement with the 360-video stemmed from the experience of the performance as a whole, including the interactions with the performers and other audience members, and the audience participation in the performance. The concept of frictions has drawn attention to:

- the audience' emotional engagement with the story, and the possibility enabled at a discursive level to share their own memories;

- the tension between the audience' perception of reality, and the fictional reality of the performance;
- the tension between the audience sense of presence in the 360-video and the sensorial experience of one the performers holding their hands.

While 360-degree videos are conceived to provide immersive experiences [46], we argue that the qualities of this immersiveness are not given, but emergent and tightly connected to frictions and to the audience's transition towards a more active participation in the performance.

Frictions in Priming and Participation

Previous work [30] has described "behind the scenes activities" to engage participants with the performance, and to prepare them to be part of the experience. As we have seen in our analysis, temporal frictions can be regarded as invitations for the audience to transition towards a more active role in the performance. This reciprocal transaction of stories and memories primes the audience members while adding meaning to the 360-degree video and contributing to the audience engagement with the performance as a whole. Thus, drawing attention to frictions illustrates how priming can be a central element of the story and of the Mixed-Reality Performance, rather than confined to a set of behind the scene activities.

The experience of the temporal friction contributed to the audience's transition between *re-telling* and *re-living* a memory. These two modes of experiences were held together by the fact that re-telling primed the re-living, an aspect of the performance that the audience members became aware of. Similarly, the frictions between the physical and virtual presence blurred the boundaries between *inhabiting* the physical and the virtual spaces of the live performance and *being present* in the narration. The audience members associated the experience of this friction to a more central role in the performance as they became characters in the story rather than just witnessing it. Finally, the friction between the fictional reality and the audience's experience of the actual world redefines participation as characterised by *believing* and *projecting*. Both believing and projecting tightly relate to the concept of suspension of disbelief [20]. Believing in our case can be explained in terms of traditional narratology studies of narratives such as books, movies or theatre plays: the audience accepts to believe in a story containing fictive elements. Projecting, however, more tightly relates to the audience's empathy with the performance, and was intertwined with the digital media used. As seen in the analysis, for some audience members this projection was in fact hindered by technical qualities of the technology, and by concerns on the role such type of media might have in the future.

Designing Immersion Through Frictions

Strange Days, as the artists themselves explained, was designed and performed using the Brechtian method of acting [39] to create a dialogue and interaction between the performer and the audience within the performance. This form of directly addressing the audience is what James Mcteague [39] considers *"breaking the fourth wall"*. In the second room however, when the performer asked her audience members to put on HMDs to take her place in the 360-degree video simulating her memory, the audience's role shifts from active and engaged towards a more passive and less interactive one. In fact, the audience became the 'observer' of the performance rather than making any interactions with the performer. This lack of interaction–and limited opportunity for action–with technology or a live performance can disturb the audience's suspension of disbelief (belief or projection). Experiences that rely on the technology to make the audience feel immersed are most susceptible to this disruption [41]. However, by establishing an embodied interaction between the performer and audience the experience became once again interactive, tangible and kinesthetic through friction. This bodily interaction, actuated by a human agent with tactile input, not only broke the forth wall, but also recreated the audience engagement with the space around.

Many strategies for creating immersive experiences, including breaking the fourth wall, have been used in the design of MRPs (i.e., *Situation Rooms* [61]). However, many performances that incorporate 360-degree video or VR do not exploit the full potential of such strategies because of the presumed "immersiveness" of the technology itself. The "successful" immersive experience in such performances are not far from experiencing, for example, the Circle-Vision theatre installed by Walt Disney Company in their theme parks, in late 60s. The main difference between this early 360-degree projection room (and the VR caves that followed it) and the types of experience discussed in this paper is the use of a headset. A Head-Mounted Display enables a more intimate experience, but it also isolates audience members from one another. Creating an immersive experience in MRPs however, does not need to rely on the technology or immersive media intrinsically. Immersion can emerge from props and technologies, how they are designed and used by the artists in the MRP, the audience's spatial and emotional engagement with the scene, with the narratives, and with the characters as performed by artists and performers. Immersion is, therefore, an emerging quality of the performance enabled by the interactions between the performer and the audience.

Reflecting on the role of frictions can lead to design strategies that aim to break the fourth wall, thus contributing to the audience transition towards an active role in the performance, and, ultimately, an immersive experience. New technological improvement in VR equipment (i.e., VR motion platforms or haptic controllers) are driven by similar goals–to create subtle connections between realities and their associated characters. This can be seen in those VR devices that add a layer of AR to the experience (i.e., Microsoft HoloLens) or in a more novel way by breaking the fourth wall with "merging the realities" as we can see in, for example, Intel's Project Alloy [60] or Facebook's Social VR

[62]. Adopting the notion of frictions provide us with an analytical lens to investigate such upcoming VR experiences, and opportunities to design interactive and embodied experiences for and with them.

Frictions in the Wild

While performance art has made inroads into HCI research, notably with works presented in [7,48,50], there are also opportunities for HCI to learn more from "real" world productions and how practitioners use different methods and technologies in their art piece.

While frictions are designed and enacted by the artists, they are experienced by the audience through participation in the performance. It is the "live" experience of frictions that creates them. The practitioners use frictions to manipulate audiences' perception of time and space, through the "live" experience and in order to create different worlds and realities in an immersive experience. These frictional realities can twist, tangle and sometimes unravel but they always connect again and come together, as they present an immersive experience. In these regards, we argue that an engaging successful immersive Mixed-Reality Performance does not merely stem from the experience of the immersive technology or media. It emerges instead from the intertwinement of the **technology** together with other dimensions, namely: the **audience's participation and contribution** to the performance and different types of **frictions** enacted by the artists to contribute to the audience's experience of the performance as a cohesive narration and frictional reality.

CONCLUSION

In this paper, we have presented the analysis of a Mixed-Reality Performance, *Strange Days*. In our analysis, we retrospectively applied the lens of frictions to better understand how artists created a more emotionally and physically immersive experience for the audience, and to better understand the audience members' experience. We use this MRP as an example of increasing interest in using 360-degree videos in artistic productions. Such performances, we have argued, do not need to rely on immersive technology to be considered as a fully immersive experience, but instead can draw on the frictions between different layers of design to prime the audience members, and engage the audience with the performance, enabling immersion with the 360-degree video and the performance itself.

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